INCH-POUND

MIL-PRF-44394A
9 February 1996
SUPERSEDING
MIL-B-44394(GL)
29 December 1989

PERFORMANCE SPECIFICATION

BEEF SNACKS, STRIPS, CURED,

PACKAGED IN A FLEXIBLE POUCH, SHELF STABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance criteria for production of a finished product consisting of shelf stable beef snack strips, cured, packaged in a flexible pouch, suitable for use by the Department of Defense as a component of operational rations. This component is consumed by combat personnel under worldwide environmental extremes as part of an operational ration, and is a source of nutritional intake. It is essential that this item be produced in accordance with good commercial practice to attain high standards of appearance, odor, flavor, and texture so that high levels of troop acceptance are achieved.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Soldier Systems Command, Natick Research, Development, and Engineering Center, ATTN: SSCNC-WRE, Natick, MA 01760-5018 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 4 of this specification, whether or not they are listed.

2.2 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.1).

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC International

(Application for copies should be addressed to the AOAC International, 481, North Frederick Avenue, Suite 500, Gaithersburg, MD 20877.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.1), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Performance characteristics of the packaged food.

3.2.1 Shelf life. The packaged food shall meet the minimum shelf life requirement of 36 months at 80°F or 6 months at 100°F.

3.2.2 Appearance.

3.2.2.1 Beef snack strips. The cured beef snack strips shall be free of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, and shall have a characteristic dark reddish brown to dark brown color. The intact beef snack strips shall measure 0.33 (± 0.06) inches thick, 0.92 (± 0.13) inches wide, and 5.7 (± 0.3) inches long.

3.2.2.2 General. There shall be no bone or bone fragment in the packaged food.

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3.2.3 Odor and flavor.

3.2.3.1 General. The packaged food shall have a well blended, slightly smoked, slightly spicy odor and flavor.

3.2.3.2 Foreign. The packaged food shall be free from foreign odors and flavors such as, but not limited to, burnt, scorched, rancid, sour, or stale.

3.2.4 Texture. The beef snack strips shall be firm and chewy.

3.2.5 Weight. The net weight of a commercial unit pack of beef snack strips shall be not less than 0.80 ounce.

3.2.6 Palatability. The finished product shall be equal to or better than the approved first article in palatability and overall appearance.

3.2.7 Water activity. The water activity (Aw) value for the beef snack strips shall be not greater than 0.810.

3.2.8 pH. The pH of the beef snack strips shall be not greater than 6.0.

3.2.9 Microbiological counts.

3.2.9.1 Aerobic plate count. The aerobic plate count for the beef snack strips shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

3.2.9.2 E. coli count. The E. coli count shall be less than 3 per gram (no positive tubes in the standard 3 tube MPN technique) in 4 of 5 samples and not greater than 10 per gram in any sample.

3.2.10 Nutrient content.

3.2.10.1 Water content. The water content of an individual pouch shall be not greater than 48.5 percent.

3.2.10.2 Protein content. The protein content of an individual pouch shall be not less than 28.2 percent.

3.2.10.3 Fat content. The fat content of an individual pouch shall be not greater than 15.8 percent.

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3.2.10.4 Ash content. The ash content of an individual pouch shall be not greater than 9.1 percent.

3.2.10.5 Salt content. The salt content of an individual pouch shall be not greater than 6.6 percent.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article shall be inspected in accordance with the provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the performance characteristics or any appearance or palatability failure shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the methods of inspection of 4.5.1 through 4.5.10.

4.4 Examination. The finished product shall be examined for compliance with the requirements specified in 3.2.2, 3.2.3, 3.2.4, 3.2.5, and 3.2.6. Defects and defect classifications are listed in table I.

TABLE I. Product defects 1/ 2/

Category		Defect
Major	Minor	
		Appearance
101		Beef snack strips not intact
102		Color not characteristic of cured beef snack strips
	201	Dimensions of beef snack strips not as specified
	202	Presence of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material

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TABLE I. Product defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
		Odor and flavor
103		Odor and flavor not as specified
		Texture
	203	Texture of beef snack strips not as specified
		Weight
	204	Net weight not as specified

1/ Presence of any foreign odors and flavors such as, but not limited to burnt, scorched, rancid, sour, or stale shall be cause for rejection of the lot.

2/ Finished product not equal to or better than the approved first article in palatability and overall appearance shall be cause for rejection of the lot.

4.5 Methods of inspection.

4.5.1 Shelf life. Verification of shelf life shall be by contractor's certificate of conformance.

4.5.2 Appearance. Appearance is determined by compliance with 4.4.

4.5.3 Odor and flavor. Odor and flavor is determined by compliance with 4.4.

4.5.4 Texture. Texture is determined by compliance with 4.4.

4.5.5 Weight. The net weight of the commercial unit pack shall be determined by weighing each sample unit on a suitable scale. Results shall be reported to the nearest 0.01 ounce.

4.5.6 Palatability. Palatability is determined by compliance with 4.4.

4.5.7 Water activity (Aw). The samples shall be prepared and analyzed for water activity in accordance with the method of the Official Methods of Analysis of the AOAC. The Aw shall be measured at 25°C. Test results shall be reported to the nearest 0.001 value. Any result not conforming to the Aw requirement shall be cause for rejection of the lot.

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4.5.8 pH. The samples shall be prepared and analyzed for pH in accordance with the method of the Official Methods of Analysis of the AOAC. The pH shall be measured at 25°C. Test results shall be reported to the nearest 0.1 unit. Any result not conforming to the pH requirement shall be cause for rejection of the lot.

4.5.9 Microbiological counts. The samples shall be individually tested for aerobic plate count and E. coli count in accordance with the methods of the Official Methods of Analysis of the AOAC. Any result not conforming to the aerobic plate and E. coli requirements shall be cause for rejection of the lot.

4.5.10 Nutrient content. The samples shall be prepared and analyzed for water content, protein content, fat content, ash content, and salt content in accordance with the methods of the Official Methods of Analysis of the AOAC. Test results shall be reported to the nearest 0.1 percent. Any result not conforming to water, protein, fat, ash, and salt requirements shall be cause for rejection of the lot.

5. PACKAGING

5.1 Packaging. Preservation, packaging, packing, labeling, and case marking shall be as specified in the solicitation, contract, or purchase order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. When a first article is not required (see 3.1).

6.2 Ingredients. Ingredients may be as follows: Beef, salt, flavorings, hydrolyzed soy and corn protein, monosodium glutamate, natural smoke and grill flavor, sugar, dextrose, yeast extract, caramel color, carboxymethyl cellulose, sodium nitrite, and malic acid.

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6.3 Subject term (key word) listing.

Operational ration
Food processing
Pouched food

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 35

Preparing activity:

Army - GL

(Project 8940-0844)

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-44394A	2. DOCUMENT DATE (*YYMMDD*) 1996 February 9

3. DOCUMENT TITLE BEEF SNACKS, STRIPS, CURED, PACKAGED IN A FLEXIBLE POUCH, SHELF STABLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME U.S. Army Soldier Systems Command, Natick Research, Development, and Engineering Center	b. TELEPHONE *(Include Area Code)* (1) Commercial 508-233-5907 (2) AUTOVON/DSN 256-5907
c. ADDRESS *(Include Zip Code)* Commander, U.S. Army Soldier Systems Command, Natick Research, Development, and Engineering Center ATTN: SSCNC-WRE, Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: **Standardization Program Division** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466** **Telephone (703) 681-9340 AUTOVON 761-9340**

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